

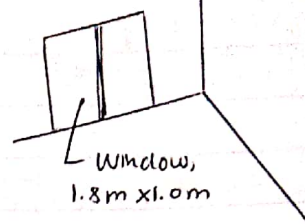
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CHE 312 - Homework #8

9.15 height 1.8m, width 1.0m
room wall and air temps = 15°C



(a) The window would show a frost layer at the base rather than at the top since heat rises, and the bottom is thinner than the top so the bottom would be colder.

(b) rate of heat loss through the window due to free convection and radiation. window has uniform temp of 0°C and $\epsilon = 0.94$
utility rate of 0.18 \$/(kW)·h (estimate corresponding daily cost)
assumptions: steady state, vertical surface = free convection, and the window has uniform temperature

At: $T_f = \frac{T_s + T_\infty}{2} = \frac{15}{2} = 7.5^\circ\text{C} = 280.65\text{ K}$ and 1 atm

Table A-4 $\nu = 14.17 \times 10^{-6} \text{ m}^2/\text{s}$
 $k = 0.0248 \text{ W/mK}$
 $\alpha = 1.995 \times 10^{-5} \text{ m}^2/\text{s}$
 $Pr = 0.712$

$q_{\text{loss}} = q_{\text{cv}} + q_{\text{rad}} \dots (1)$

$q_{\text{loss}} = A_s [\bar{h}_L (T_\infty - T_s) + \epsilon \sigma (T_{\text{sur}}^4 - T_s^4)] \dots (2)$

$\overline{Nu}_L = \frac{\bar{h}_L L}{k} = \left\{ 0.825 + \frac{0.387 Ra_L^{1/4}}{[1 + (0.492/Pr)^{1/4}]^{1/4}} \right\}^2 \dots (3)$

$Ra_L = Gr_L Pr = \frac{g \beta (T_s - T_\infty) L^3}{\nu \alpha} \approx 10^9 \dots (4)$
 $\beta = \frac{1}{T_f}$

$Ra_L = \frac{(9.8 \text{ m/s}^2)(3.57 \times 10^{-3} \text{ 1/K})(15 - 0)(1.8)^3}{(14.17 \times 10^{-6} \text{ m}^2/\text{s})(1.995 \times 10^{-5} \text{ m}^2/\text{s})} = 1.083 \times 10^{10}$

$\overline{Nu}_L = \left\{ 0.825 + \frac{0.387(1.083 \times 10^{10})^{1/4}}{[1 + (0.492/0.712)^{1/4}]^{1/4}} \right\}^2 = 258.8$

$$\overline{Nu}_L = \frac{\overline{h}_L L}{k} \Rightarrow \frac{\overline{Nu}_L k}{L} = \overline{h}_L = \frac{(258.8)(0.0248 \text{ W/mK})}{(1.8)} = 3.56 \text{ W/m}^2\text{K}$$

$$q_{\text{loss}} = A_s [\overline{h}_L (T_{\infty} - T_s) + \epsilon \sigma (T_{\text{surr}}^4 - T_s^4)]$$

$$q_{\text{loss}} = (1 \times 1.8) [(3.56 \text{ W/m}^2\text{K})(15) + (0.94)(5.67 \times 10^{-8})(288.15^4 - 273.15^4)]$$

$$q_{\text{loss}} = (1.8 \text{ m}^2) [53.4 \text{ W/m}^2 + 70.74 \text{ W/m}^2]$$

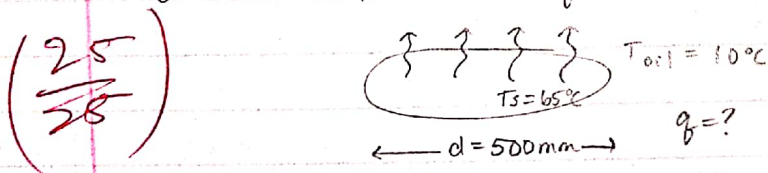
$$q_{\text{loss}} = 223.45 \text{ W} = 0.22345 \text{ kW}$$

The daily cost : $q_{\text{loss}} (\text{utility rate}) (24 \text{ hrs})$

$$= (0.22345 \text{ kW})(0.18 \text{ \$/kW} \cdot \text{hr})(24 \text{ hrs})$$

$$= \boxed{0.965 \text{ \$/day}}$$

9.28 an electrical heater $\rightarrow$ horizontal disk of 500-mm diameter used to heat the bottom of a tank filled w/ engine oil at $T = 10^\circ\text{C}$ calculate power required to maintain the heater surf. temp. at 65°C



Assumptions: ambient air is quiescent, surface radiation effects are neglectable, and constant properties

$$T_f = \frac{T_s + T_{\infty}}{2} = \frac{65 + 10}{2} = \frac{75}{2} = 37.5^\circ\text{C} = 310.65 \text{ K}$$

Table A.5: engine oil (unused)

$$\nu = 288 \times 10^{-6} \text{ m}^2/\text{s}$$

$$k = 0.145 \text{ W/m} \cdot \text{K}$$

$$\alpha = 0.847 \times 10^{-7} \text{ m}^2/\text{s}$$

$$Pr = 3400$$

$$\beta = 0.70 \times 10^{-3} \text{ 1/K}$$

$$\frac{\pi r^2}{2\pi r} = \frac{r}{2}$$

$$L = \frac{A_s}{P} = \frac{\frac{\pi D^2}{4}}{\pi D} = \frac{D}{4} = \frac{0.5 \text{ m}}{4} = 0.125 \text{ m}$$

$$Ra_L = \frac{g \beta (T_s - T_\infty) L^3}{\nu \alpha} = \frac{(9.8 \text{ m/s}^2) (0.70 \times 10^{-3} \text{ K}) (65 - 10^\circ\text{C}) (0.125 \text{ m})^3}{(288 \times 10^{-6} \text{ m}^2/\text{s}) (0.847 \times 10^{-7} \text{ m}^2/\text{s})}$$

$$Ra_L = 3.02 \times 10^7$$

$$\overline{Nu}_L = 0.15 Ra_L^{1/3} \quad (10^4 \leq Ra_L \leq 10^7)$$

$$\overline{Nu}_L = 0.15 (3.02 \times 10^7)^{1/3} = 46.71$$

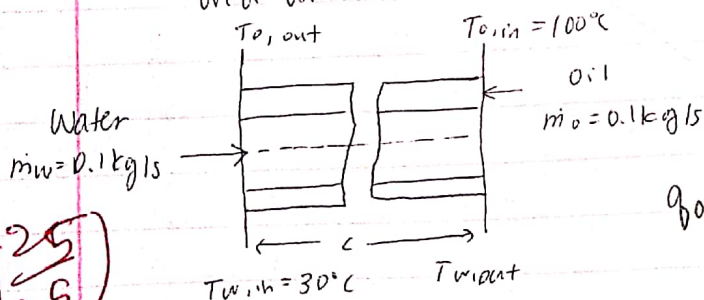
$$\overline{Nu}_L = \frac{\bar{h}_L L}{k} \rightarrow \bar{h}_L = \frac{\overline{Nu}_L k}{L} = \frac{(46.71) (0.145)}{(0.1)} = 54.18 \text{ W/m}^2\text{K}$$

$$q = \bar{h}_L A_s (T_s - T_\infty) = \bar{h}_L \left(\frac{\pi D^2}{4} \right) (T_s - T_\infty)$$

$$q = (54.18 \text{ W/m}^2\text{K}) \left(\frac{\pi (0.5 \text{ m})^2}{4} \right) (65 - 10)$$

$$q = \boxed{588.10 \text{ W}}$$

11-20 thin-walled inner tube of 25-mm diameter carrying water and an outer tube of 25-mm diameter carrying the oil. $55 \text{ W/m}^2\text{K}$



(a) outlet temp is 50°C

$$q_{\text{oil}} = \dot{m}_o c_{p,o} (T_{o,i} - T_{o,o}) = (0.1 \text{ kg/s}) (1900 \text{ J/kg}\cdot\text{K}) (100^\circ\text{C} - 50^\circ\text{C}) = \boxed{9500 \text{ J}}$$

$$q_{\text{water}} = 9500 = (0.1 \text{ kg/s}) (4200) (T_{w,o} - 30^\circ\text{C}) \rightarrow \boxed{T_{w,o} = 52.62^\circ\text{C}}$$

b. $q = UA \Delta T_{\text{lm}}$ where $A = \pi D_i L$

$$\Delta T_{\text{lm}} = \frac{(T_{h,i} - T_{c,o}) - (T_{h,o} - T_{c,i})}{\ln \left[\frac{(T_{h,i} - T_{c,o})}{(T_{h,o} - T_{c,i})} \right]} = \frac{(100^\circ\text{C} - 52.62^\circ\text{C}) - (50^\circ\text{C} - 30^\circ\text{C})}{\ln \left[\frac{(100 - 52.62)}{(50 - 30)} \right]}$$

$$\Delta T_{\text{lm}} = 31.75^\circ\text{C or K}$$

$$U = \frac{1}{\left(\frac{1}{h_i} \right) + \left(\frac{1}{h_o} \right)} = 55 \text{ W/m}^2\text{K}$$

$$L = \frac{q}{U \pi D_i \Delta T_{\text{lm}}} = \frac{(9500 \text{ J/s})}{(55 \text{ W/m}^2\text{K}) (\pi) (0.025 \text{ m}) (31.75^\circ\text{C})}$$

$$L = \boxed{69.3 \text{ m}}$$

$$11.26 \quad \epsilon \sim \dot{m}_{air}^{-0.25}$$

(a) $\dot{m}_{air}$, percent increase in the heat
 $\dot{m}_{water} > \dot{m}_{air}$
 $C_{min} = C_c = \dot{m}_{air} C_{p,c}$

$$q = \epsilon q_{max} = \epsilon C_{min} (T_{h,i} - T_{c,i}) \\ = \epsilon \dot{m}_{air} C_{p,c} (T_{h,i} - T_{c,i})$$

$$\left(\frac{28}{25} \right) \frac{q_{high}}{q_{low}} = \frac{(\epsilon \dot{m}_{air}^{0.75})_{high}}{(\epsilon \dot{m}_{air}^{0.75})_{low}} = \frac{2^{0.75}}{1} = 1.682$$

$$\text{Percentage increase} = \frac{1.682 - 1}{1} = \boxed{68.2\%}$$

(b) low-speed fan condition, outdoor air from $-10 \pm 0.3^\circ\text{C}$, airflow increases 50%.
 Find the new outlet temp.

$$\frac{q_{med}}{q_{low}} = 1.5^{0.75} = 1.355 \frac{(\dot{m}_{air})_{med}}{(\dot{m}_{air})_{low}} = 1.5$$

$$\frac{q_{med}}{q_{low}} = \frac{[\dot{m}_{air} C_{p,c} (T_{c,o} - T_{c,i})]_{med}}{[\dot{m}_{air} C_{p,c} (T_{c,o} - T_{c,i})]_{low}}$$

$$1.355 = \frac{1.5 [T_{c,o} - (-10^\circ\text{C})]}{[30 - (-10)]} \rightarrow T_{c,o} = \boxed{26.14^\circ\text{C}}$$